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EEL 3135 – Classwork #04

Milestone:

- Complete any 1 question and 2 parts of a second question to receive credit.

Question #1: Consider

$$x(t) = 6 + 10 \cos\left(240\pi t - \frac{\pi}{6}\right) + 5 \cos\left(360\pi t + \frac{\pi}{4}\right).$$

- (a) Convert each cosine's angular frequency to Hz and list f_1, f_2 . Determine if $x(t)$ is periodic by checking if $f_1 = K_1 f_0$ and $f_2 = K_2 f_0$ for integers K_1, K_2 with $\gcd(K_1, K_2) = 1$. Report f_0, ω_0, T_0 .

$$240\pi \rightarrow 120\text{Hz}, 360\pi \rightarrow 180\text{Hz}$$

$$\text{GCD}(120, 180) = 60 \text{ Hz}$$

$$f_0 = 60\text{Hz}, \omega_0 = 120\pi, T_0 = \frac{1}{60} \text{ seconds}.$$

$$120 = 60 * 2, \text{ where } 2 \text{ is our } k \text{ value}$$

$$180 = 60 * 3, \text{ where } 3 \text{ is our } k \text{ value}$$

$$\text{GCD}(2, 3) = 1 \text{ 😊}$$

- (b) Append a third term $4\cos(600\pi t)$. Re-test periodicity and recompute f_0, ω_0, T_0 if periodic. Comment on how adding terms can break or restore periodicity.

$$\text{GCD}(120, 180, 300) = 60 \text{ Hz}$$

$$f_0 = 60\text{Hz}, \omega_0 = 120\pi, T_0 = \frac{1}{60} \text{ seconds}.$$

$$120 = 60 * 2$$

$$180 = 60 * 3$$

$$300 = 60 * 5$$

$$\text{GCD}(2, 3, 5) = 1 \text{ 😊}$$

- (c) **MATLAB:** Write a script that:

- Defines $x(t)$ over $t \in [0, 4T_0]$ with $f_s = 20\text{kHz}$,
- Plots $x(t)$ and visually verifies periodicity over multiple periods (add grid, labels),
- Overlays vertical lines every T_0 .

- (d) Briefly explain why choosing a smaller valid f_0 (when possible) can change a “non-periodic” result to a periodic one.

Question #2: Sum/Product & Amplitude Modulation

Let $f_c = 400 \text{ Hz}$ and $\Delta = 20 \text{ Hz}$:

$$x_{\text{sum}}(t) = \cos(2\pi(f_c - \Delta)t) + \cos(2\pi(f_c + \Delta)t).$$

- (a) Use complex exponentials to show:

$$x_{\text{sum}}(t) = 2\cos(2\pi\Delta t)\cos(2\pi f_c t).$$

Then state f_0 conditions based on $\gcd(K-1, K+1)$ where $f_c = K\Delta$.

- (b) **MATLAB:** Plot $x_{\text{sum}}(t)$ for $t \in [0, 0.5]$ s with $f_s = 10\text{kHz}$. In separate figures, plot $\cos(2\pi\Delta t)$ and $\cos(2\pi f_c t)$. Label clearly and add legends.
- (c) Let $v(t) = 3 + 2.5\cos(60\pi t)$ and $x(t) = v(t)\cos(2\pi \cdot 150t)$. Expand $x(t)$ into cosines (carrier + sidebands) and sketch/label the spectrum.
- (d) **MATLAB:** Generate $x(t)$ numerically and stem-plot its spectrum. Verify amplitudes match your algebra.

Question #3: Orthogonality & Reconstruction

Consider the periodic signal

$$x(t) = 4 + 6 \cos\left(\frac{\pi}{4}t\right) + 3 \sin\left(\frac{\pi}{3}t\right).$$

- (a) Rewrite $x(t)$ using inverse Euler to express $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$, identify ω_0 , and list the nonzero a_k .

- (b) State the Fourier series coefficient definition:

$$a_k = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-jk\omega_0 t} dt$$

and explain why orthogonality justifies this extraction.

- (c) **MATLAB:** Compute a_k numerically for $|k| \leq 6$ using trapz over one period. Display a table of k vs. a_k (real & imag parts). Compare with your analytic values.
- (d) **MATLAB:** Make a two-sided stem plot of $|a_k|$ for $|k| \leq 6$. Clearly label the nonzero coefficients.
- (e) Comment briefly on why coefficients at unused harmonics are ≈ 0 (numerical orthogonality).

Question #4: Fourier Coefficients, Spectrum & Reconstruction

Consider the periodic signal

$$x(t) = 5 + 6 \cos\left(\frac{\pi}{5}t - \frac{\pi}{6}\right) - 4 \cos\left(\frac{\pi}{3}t + \frac{\pi}{4}\right) + 3 \sin\left(\frac{\pi}{2}t\right).$$

(a) Express $x(t)$ as a Fourier Series:

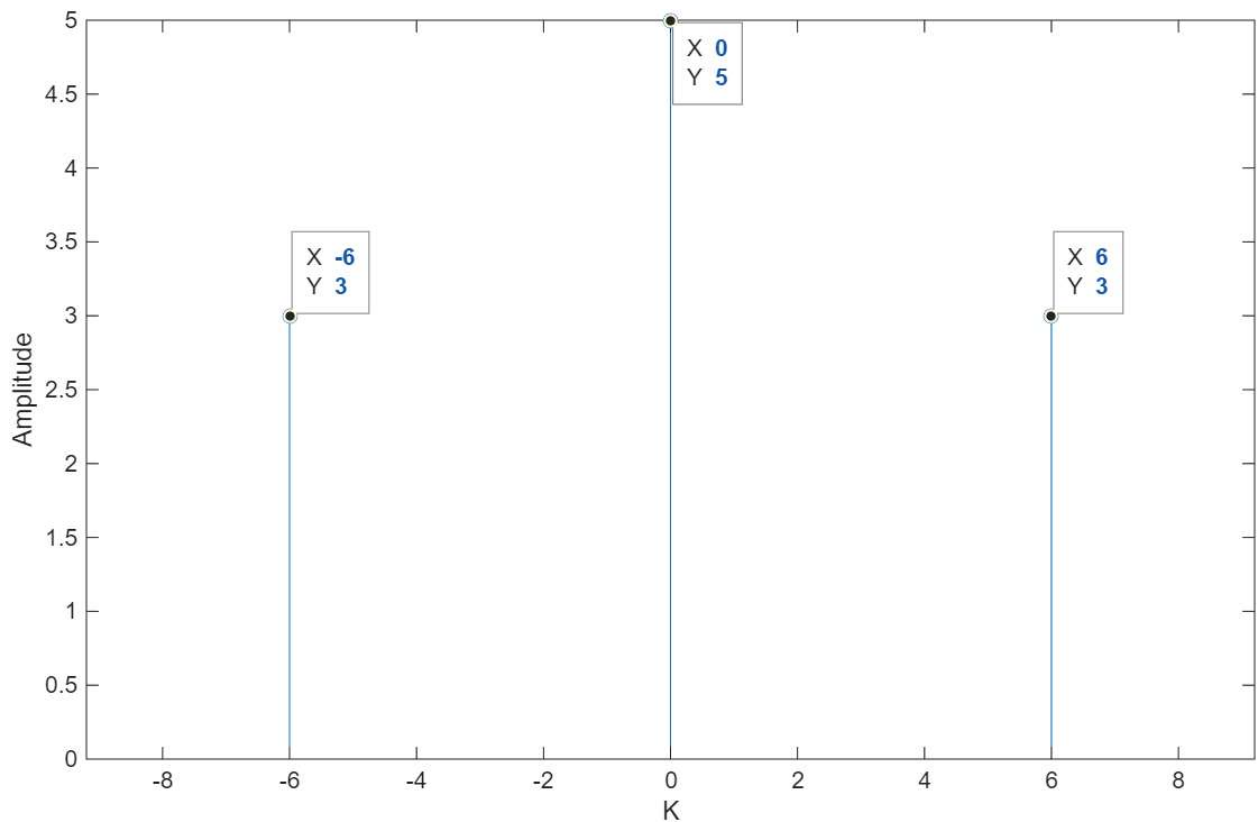
$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Compute all nonzero a_k values by rewriting each term using inverse Euler's formula and combining terms.

$$\text{Fundamental Frequencies: } 0, \frac{\pi}{5}, \frac{\pi}{3}, \frac{\pi}{2}$$

- GCD: $\frac{\pi}{30}$
- $\frac{\pi}{5} = \frac{\pi}{30} * 6$
- $\frac{\pi}{3} = \frac{\pi}{30} * 10$
- $\frac{\pi}{2} = \frac{\pi}{30} * 15$
- $K = -15, -10, -6, 0, 6, 10, 15.$
- $6 \cos\left(\frac{\pi}{5}t - \frac{\pi}{6}\right)$ Inverse Euler's:
 - $= 3e^{j\frac{\pi}{5}t}e^{-j\frac{\pi}{6}} + 3e^{-j\frac{\pi}{5}t}e^{j\frac{\pi}{6}} \rightarrow a_{\pm 6}$
- $-4 \cos\left(\frac{\pi}{3}t + \frac{\pi}{4}\right)$ Inverse Euler's:
 - $= -2e^{j\frac{\pi}{3}t}e^{j\frac{\pi}{4}} - 2e^{-j\frac{\pi}{3}t}e^{-j\frac{\pi}{4}} \rightarrow a_{\pm 10}$
- $3 \sin\left(\frac{\pi}{2}t\right)$ Inverse Euler's:
 - $= -j1.5e^{j\frac{\pi}{2}t} - j1.5e^{-j\frac{\pi}{2}t} \rightarrow a_{\pm 15}$

- (b) MATLAB: Create a two-sided stem plot of $|a_k|$ vs. k for $|k| \leq 6$. Clearly mark k values where $|a_k| = 0$.



- (c) Using the a_k coefficients, reconstruct an approximation of $x(t)$:

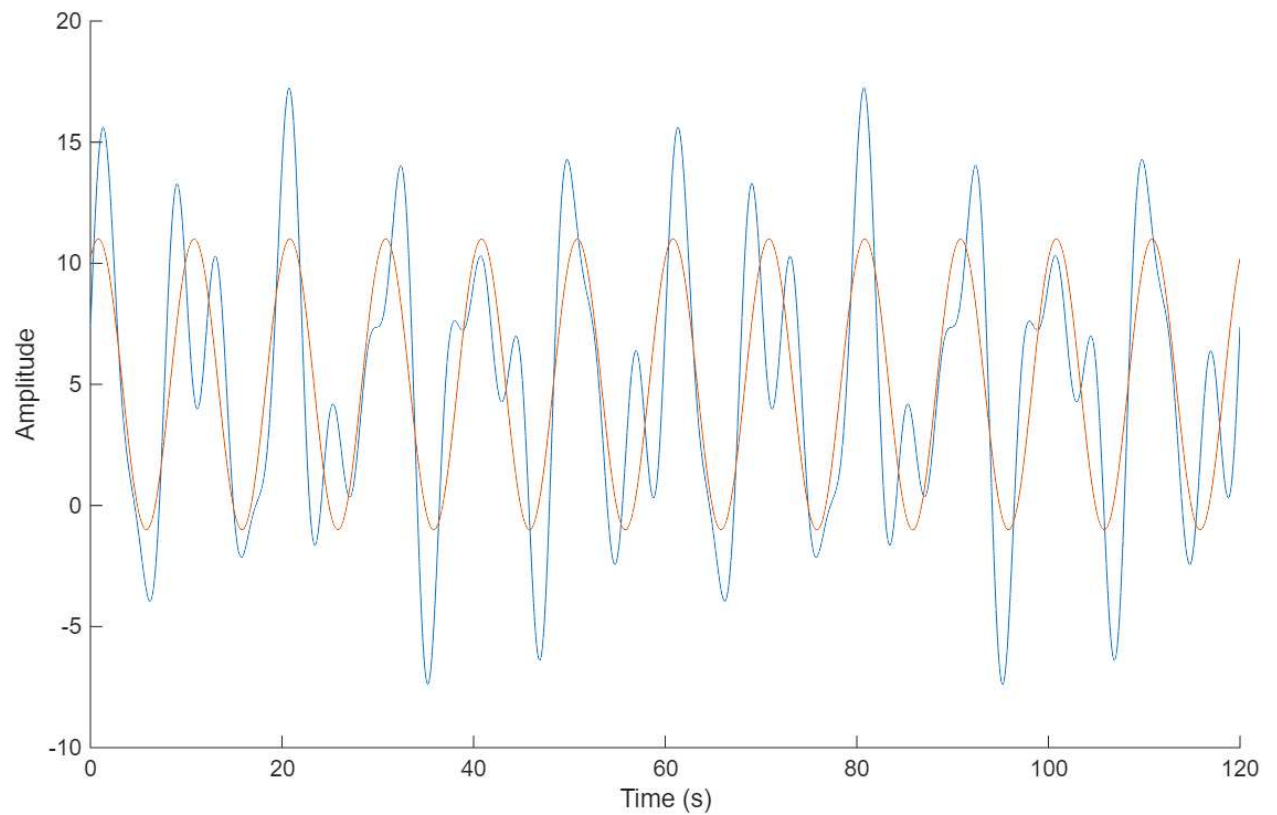
$$\hat{x}(t) = \sum_{k=-6}^6 a_k e^{jk\omega_0 t}$$

$$x_1 = 5 + 6 \cos\left(\frac{\pi}{5}t - \frac{\pi}{6}\right) - 4 \cos\left(\frac{\pi}{3}t + \frac{\pi}{4}\right) + 3 \sin\left(\frac{\pi}{2}t\right)$$

$$x_2 = 5 + 6 \cos\left(\frac{\pi}{5}t - \frac{\pi}{6}\right)$$

$$\text{Fundamental frequency} = \frac{\pi}{30}$$

Plot both $x(t)$ and $\hat{x}(t)$ on the same graph over $t \in [0, 2T_0]$ and comment on how well they match.



(d) Construct sinusoids from the following Fourier coefficients a_k .

$$a_0 = 2,$$

$$a_{-1} = 3,$$

$$a_1 = 3,$$

$$a_{-4} = -4j,$$

$$a_4 = 4j \quad (\text{fundamental frequency: } \omega_0 = 40)$$

$$x(t) = 2 + 6 \cos(40t) - 8 \sin(160t)$$